Penetration Testing Report

Footprinting, Reconnaissance & Network Scanning

Week 2 W2-PM1 & W2-PM5

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Program / Batch	B084 - Networkwalks Cybersecurity Internship
Date	9 November 2026
Modules Completed	W2-PM1 (Multiple Kali Tools) -Complete | W2-PM5 (Zenmap) Complete
Target Scope	networkwalks.com (public website + DNS) and tester's own LAN, per signed Letter of Authorization NW-LOA-B082-017

Liability Disclaimer

This report documents educational, authorized security testing performed under the NetworkWalks Cybersecurity Internship (Batch B084). All activities were carried out strictly within the scope and dates authorized in writing by the Client (Networkwalks), per Letter of Authorization NW-LOA-B082-017. No exploitation, unauthorized access, or disruptive activity was attempted. Findings are recorded for educational purposes only.

Introduction

Footprinting and reconnaissance is the first phase of a penetration test. It gathers public information about a target domain ownership, hosting, technologies, DNS records, and network layout without touching the target directly, laying the foundation for later testing phases.

W2-PM1: Footprinting with Multiple Kali Tools (networkwalks.com)

Task 1: whois

Command:

```
whois networkwalks.com
```

Findings:

- Registrar: GoDaddy.com, LLC
- Creation Date: 6 Nov 2019
- Expiry Date: 6 Nov 2027
- Name Servers: NS6135.HOSTGATOR.COM, NS6136.HOSTGATOR.COM (confirms HostGator as hosting provider)
- Registrant identity hidden via Domains By Proxy

```
┌──(kali㉿kali)-[~]
└─$ whois networkwalks.com
   Domain Name: NETWORKWALKS.COM
   Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.godaddy.com
   Registrar URL: http://www.godaddy.com
   Updated Date: 2025-11-12T10:08:43Z
   Creation Date: 2019-11-06T22:51:46Z
   Registry Expiry Date: 2027-11-06T22:51:46Z
   Registrar: GoDaddy.com, LLC
   Registrar IANA ID: 146
   Registrar Abuse Contact Email: abuse@godaddy.com
   Registrar Abuse Contact Phone: 480-624-2505
   Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
   Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
   Name Server: NS6135.HOSTGATOR.COM
   Name Server: NS6136.HOSTGATOR.COM
   DNSSEC: unsigned
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-11-10T21:59:45Z <<<

For more information on Whois status codes, please visit https://icann.org/epp
```

Task 2: whatweb

Command:

```
whatweb networkwalks.com
```

Findings:

- Web server: Apache
- CMS: WordPress 7.1.4
- Plugin: WordPress Download Manager 3.3.76
- Bootstrap 7.1.4, jQuery 3.7.1
- Server IP: 192.232.216.135
- Contact email exposed: info@networkwalks.com
- Country: United States



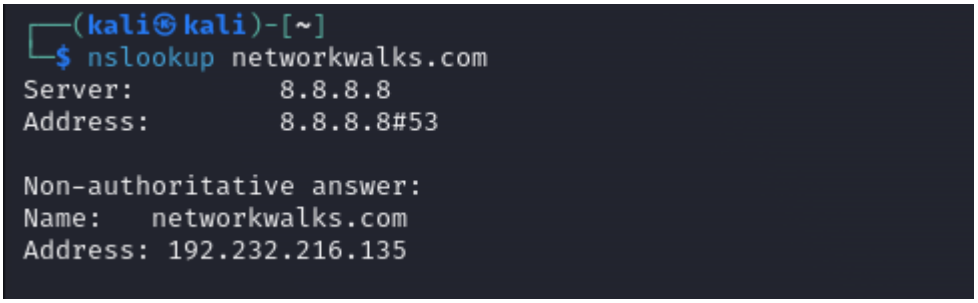
Task 3: nslookup

Command:

```
nslookup networkwalks.com
```

Findings:

- Resolves to IP: 192.232.216.135 (matches the IP found via whatweb)
- DNS server used: 8.8.8.8 (Google DNS)



Task 4: curl -I

Command:

```
curl -I https://networkwalks.com
```

Findings:

- Server: Apache
- WordPress confirmed again via x-nginx-cache header
- A WordPress REST API endpoint is exposed via the link header (/wp-json/)
- Cache-related headers (x-endurance-cache-level, x-nginx-cache) leak the caching stack
- CSP/permissions-policy headers reference reCAPTCHA, hCaptcha, and Cloudflare challenges



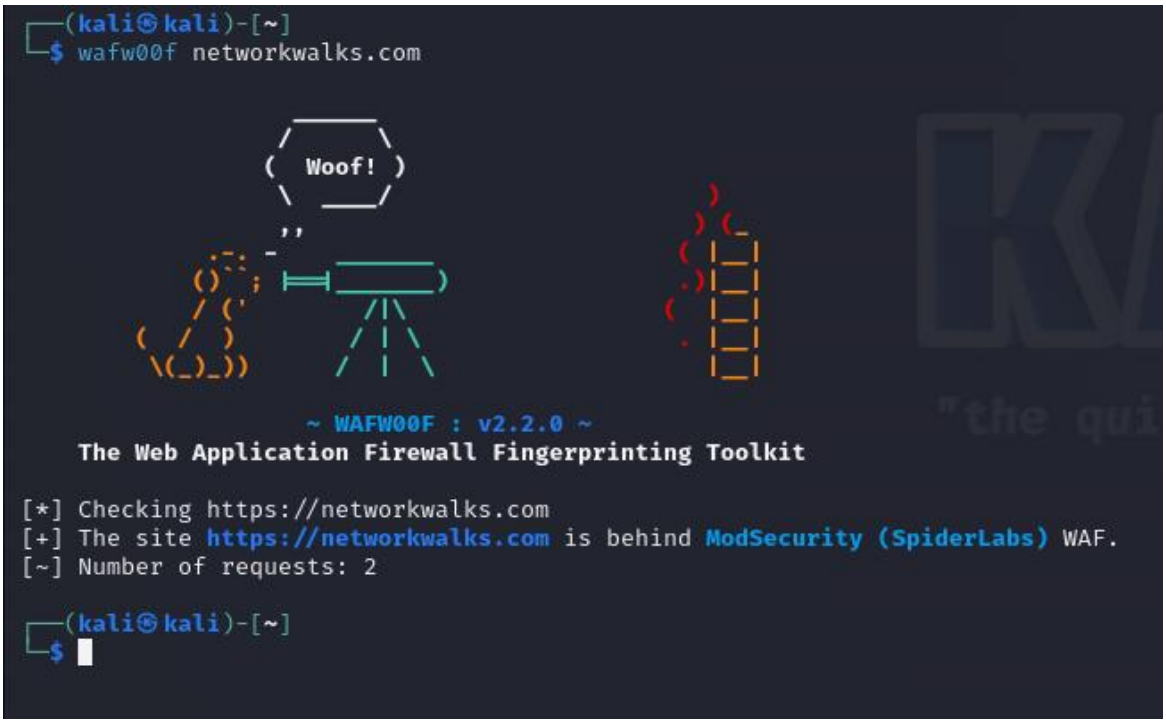
Task 5: wafw00f

Command:

```
wafw00f networkwalks.com
```

Findings:

- The site is behind a ModSecurity (SpiderLabs) Web Application Firewall
- This means naive scanning/exploitation attempts would likely be blocked or logged



Task 6: dnsrecon

Command:

```
dnsrecon -d networkwalks.com
```

Findings:

- DNSSEC not configured
- Name servers: ns6135.hostgator.com, ns6136.hostgator.com (Bind version 9.16.23-RH on both)
- Mail server (MX): mail.networkwalks.com (192.232.216.135)
- SPF record: v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
- Google site verification TXT record present
- 8 SRV records found all pointing to cPanel email autodiscovery (cpanelemaildiscovery.cpanel.net), confirming cPanel hosting



W2-PM5: Network Scanning with Zenmap/Nmap (Local LAN)

Task 1-2: Local IP and subnet

Command:

```
ip a
```

Kali VM local IP is 10.0.0.2/24, so the LAN subnet is 10.0.0.0/24.

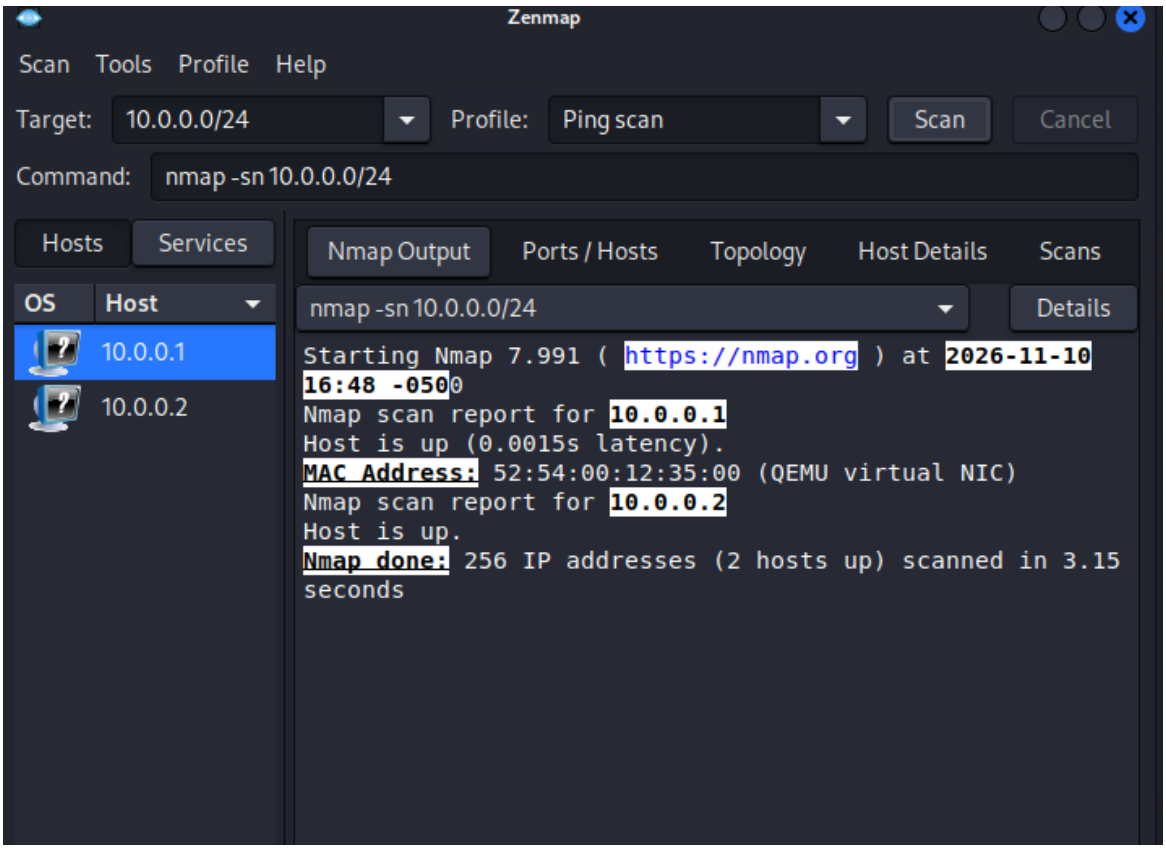
Task 3-6: Live hosts, count, IPs, MAC addresses

Command:

```
nmap -sn 10.0.0.0/24  (run via Zenmap GUI, Ping scan profile)
```

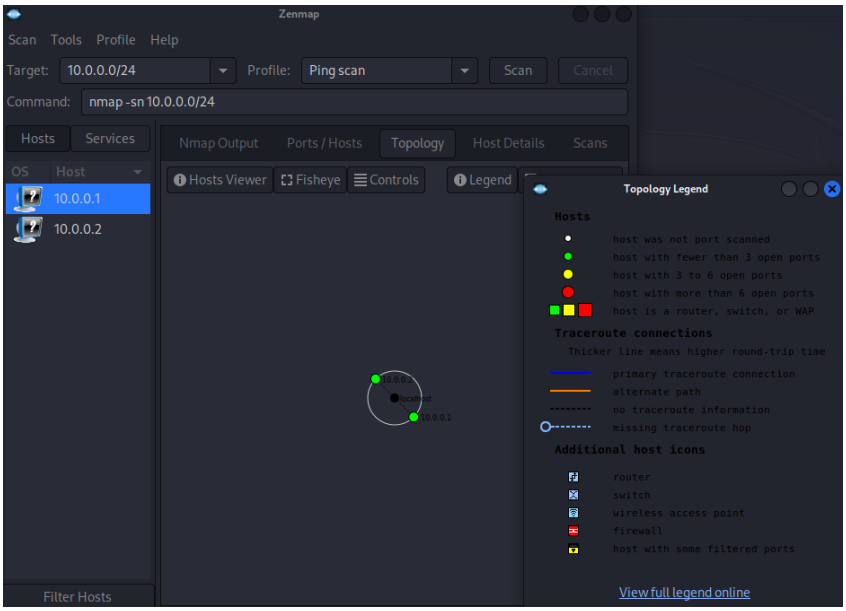
Findings:

- 2 hosts live out of 256 IP addresses scanned
- Live hosts: 10.0.0.1 (NAT gateway, MAC 52:54:00:12:35:00 QEMU virtual NIC) and 10.0.0.2 (Kali VM, this tester's machine)
- Only one other device was on the network segment (the virtual NAT gateway); the isolated lab network has no additional hosts



Task 7: Topology output

Topology view (via Zenmap GUI, Topology tab with Legend enabled) shows 2 nodes: localhost (this Kali VM) connected to the NAT gateway at 10.0.0.1. Saved as a PDF graphic for submission.


Zenmap
Scan Tools Profile Help
Target: 10.0.0.0/24
Profile: Ping scan
Scan
Cancel
Command: nmap -sn 10.0.0.0/24
Hosts
Services
OS
Host
10.0.0.1
10.0.0.2
Filter Hosts
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
Hosts Viewer
Fisheye
Controls
Legend
Topology Legend
Hosts
host was not port scanned
host with fewer than 3 open ports
host with 3 to 6 open ports
host with more than 6 open ports
host is a router, switch, or WAP
Traceroute connections
Thicker line means higher round-trip time
primary traceroute connection
alternate path
no traceroute information
missing traceroute hop
Additional host icons
router
switch
wireless access point
firewall
host with some filtered ports
View full legend online

Conclusion

This engagement demonstrated how much information about a target can be gathered using only passive, publicly-available data without ever touching the target system directly.

Footprinting (W2-PM1) on networkwalks.com revealed:

- Hosting provider (HostGator) and registrar (GoDaddy) via whois
- Exact CMS and plugin versions (WordPress 7.1.4, WordPress Download Manager 3.3.76) via whatweb both could be checked against known CVEs
- Server IP (192.232.216.135) via nslookup, confirmed across multiple tools
- Exposed REST API endpoint and caching stack via HTTP headers (curl)
- An active ModSecurity WAF via wafw00f, which would shape how an attacker approaches further testing
- Full DNS footprint (mail servers, SPF policy, cPanel SRV records) via dnsrecon, with DNSSEC notably not configured

Network scanning (W2-PM5) on the tester's own LAN confirmed host discovery fundamentals: identifying all live hosts, their IPs and MAC addresses, and visualizing the network topology skills directly transferable to scanning any authorized target network.

Taken together, these findings illustrate why organizations should minimize what they expose publicly: every piece of information gathered here (hosting provider, software versions, DNS records, WAF presence) narrows the attack surface an adversary needs to map before attempting exploitation. All testing in this report stayed within the scope and conditions of Letter of Authorization NW-LOA-B082-017.